

AI Drug Repurposing Research Report

Research Query

Research Report for Paracetamol use in
sepsis

Generated on 9/9/2026, 4:56:40 pm

Generated by Agentic AI Research Assistant

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Executive Summary

Paracetamol, also known as acetaminophen, has been explored for potential benefits in sepsis beyond its antipyretic and analgesic effects, owing to its purported anti-inflammatory and antioxidant properties. However, a significant randomized clinical trial (ASTER) demonstrated that intravenous acetaminophen did not improve the number of days alive and free of organ dysfunction in critically ill sepsis patients [PMID: 38762798]. Other concerns include acetaminophen-induced hypotension in this vulnerable population [PMID: 35505446] and an association with delayed antibiotic administration, which is critical in sepsis outcomes [PMID: 42117581]. While ongoing research investigates specific mechanisms and comparative efficacy, current direct evidence does not support a broad repurposing of paracetamol to mitigate organ dysfunction in sepsis.

Evidence Overview

The retrieved evidence includes direct literature discussing acetaminophen's role and effects in sepsis, particularly the ASTER trial. Several related literature articles cover general sepsis pathophysiology, liver failure, and potential mechanisms of acetaminophen toxicity. Clinical trial data includes a direct phase 2 trial investigating oxidative injury and several related trials comparing acetaminophen with other antipyretics, examining hemodynamic effects, or exploring toxicity in septic patients. One trial directly evaluating acetaminophen's impact on organ dysfunction in sepsis, the ASTER trial, is central to this assessment.

Evidence Database Counts (Deterministic Summary)

PubMed Publications:	DIRECT: 2 RELATED: 12 INDIRECT: 1
ClinicalTrials.gov:	DIRECT: 1 RELATED: 6 INDIRECT: 3
Total Aggregated:	DIRECT: 3 RELATED: 18 INDIRECT: 4

Literature Evidence

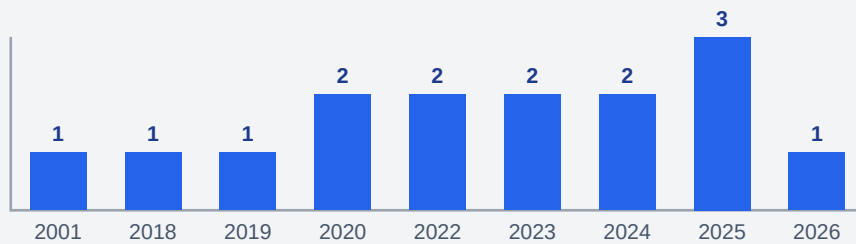
Direct literature provides crucial insights into paracetamol's utility in sepsis. The ASTER Randomized Clinical Trial investigated whether acetaminophen could increase days alive and free of organ dysfunction in critically ill patients with sepsis. The findings indicated that acetaminophen (paracetamol) did not improve these outcomes [PMID: 38762798]. Another direct study highlighted acetaminophen-induced hypotension in critically ill patients, though the role of mannitol as a stabilizing compound in formulations was also considered [PMID: 35505446].

Related literature further informs the discussion. A study explored the association between paracetamol administration and delayed intravenous antibiotic administration in hospitalized sepsis patients, suggesting that paracetamol might mask fever, a key indicator, and thus contribute to treatment delays [PMID: 42117581]. The mechanisms of acetaminophen-induced liver injury, particularly through Caspase11-dependent non-canonical pyroptosis promoted by NEDD4 lactylation, are also discussed, providing context for potential toxicity even in sepsis settings [PMID: 38385085]. While other related articles discuss general conditions like acute liver failure [PMID: 31840297, PMID: 36001996, PMID: 37288706], their direct applicability to paracetamol's repurposing in sepsis is limited, beyond general considerations for patients with liver compromise.

Retrieved PubMed Publications Database

PMID	Title	Year	Relevance
35505446	Acetaminophen-induced hypotension in sepsis.	2022	DIRECT
38762798	Acetaminophen for Prevention and Treatment of Organ ...	2024	DIRECT
40705318	Diverticulitis: A Review.	2025	RELATED
31840297	Pathophysiology of Acute Liver Failure.	2020	RELATED
33007269	Management of clinical chorioamnionitis: an evidence...	2020	RELATED
30215950	Acute Appendicitis: Efficient Diagnosis and Management.	2018	RELATED
36001996	Management of Acute Liver Failure: Update 2022.	2022	RELATED
36997396	Maternal fever in labor: etiologies, consequences, a...	2023	RELATED
38385085	NEDD4 lactylation promotes APAP induced liver injury...	2024	RELATED
37288706	Liver support systems and liver transplantation in a...	2025	RELATED
40169165	Stem cell injections for osteoarthritis of the knee.	2025	RELATED
37700323	A diagnostic model for sepsis-induced acute lung inj...	2023	RELATED
11191738	Misoprostol therapeutics revisited.	2001	RELATED
42117581	Associations between paracetamol administration and ...	2026	RELATED
30340932	PDA-TOLERATE Trial: An Exploratory Randomized Contro...	2019	INDIRECT

Publication Trend by Year (Retrieved Sample)



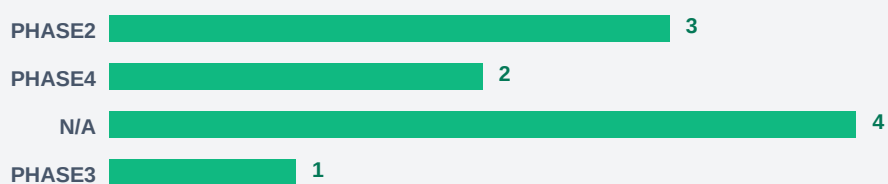
Clinical Trial Evidence

Direct clinical trial evidence includes NCT01739361,

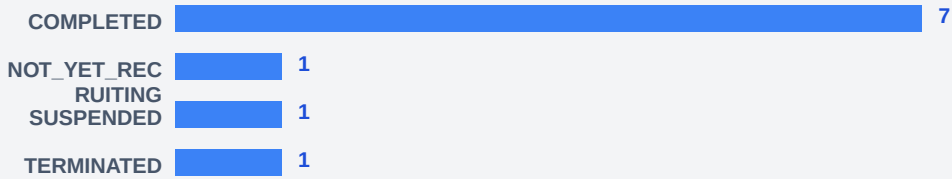
Retrieved ClinicalTrials.gov Registry Studies Database

NCT ID	Study Title	Status	Phase	Relevance
NCT01739361	Acetaminophen for the Reduction of Oxidative Injury ...	COMPLETED	PHASE2	DIRECT
NCT06061575	Comparison of the Efficacy of Paracetamol and Ibupro...	COMPLETED	PHASE4	RELATED
NCT07131969	Transcriptional Analysis of Mechanisms in Liver Fail...	NOTYETRECRUITING	N/A	RELATED
NCT01182974	Paracetamol Toxicity in Septic Patients	SUSPENDED	PHASE3	RELATED
NCT03399318	Aggressive Antipyretics for Fever Reduction in CNS M...	COMPLETED	PHASE2	RELATED
NCT06076980	Heamodynamic Effects of Paracetamol in Septic Shock ...	COMPLETED	PHASE4	RELATED
NCT02750163	Circulating Free Hemoglobin and Microcirculation Aft...	COMPLETED	N/A	RELATED
NCT04786379	Surgical Management of Knee Septic Arthritis	TERMINATED	N/A	INDIRECT
NCT05140213	A STUDY BEFORE/AFTER THE IMPLEMENTATION OF AN ANTI-I...	COMPLETED	N/A	INDIRECT
NCT05293132	Effect of Montelukast Versus Co Enzyme in Sepsis	COMPLETED	PHASE2	INDIRECT

Clinical Trials Phase Distribution (Retrieved Sample)



Clinical Trials Status Distribution (Retrieved Sample)



Drug Overview

Target Disease Overview

Repurposing Rationale

Safety & Contraindications

Repurposing Assessment

Limitations

Conclusion

References